ARROW 3 WEAPON SYSTEM

# Requirements Consistency Report

A3-INT-SSS-010 Rev A · Interceptor Segment Specification vs. the program baseline

| 11 | 4 | 6 | 1 |
|---|---|---|---|
| TOTAL FINDINGS | CRITICAL | MAJOR | MINOR |

| VERDICT | NOT READY FOR BASELINE — 4 critical and 6 major inconsistencies must be resolved before the document can enter the PDR baseline. |
|---|---|

| | |
|---|---|
| **Document under review** | A3-INT-SSS-010 Rev A — Interceptor Segment Specification |
| **Baseline checked** | 6 documents uploaded so far · 69 requirements indexed |
| **Generated** | 22 October 2026, 15:26 |
| **Produced by** | EDK custom agent “Find Inconsistencies in Requirements” (two Hermes agents) — automated analysis, to be confirmed by the Interface Working Group |
| **Classification** | UNCLASSIFIED // FICTITIOUS DEMONSTRATION DATA |

## Blocking findings

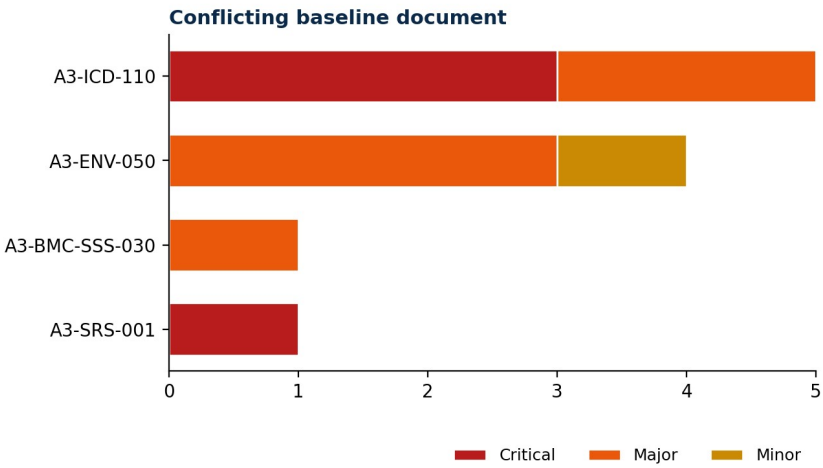
| | | |
|---|---|---|
| **F-01** | **Data Interface Mismatch** | INT-DAT-020 ↔ IF-DAT-001<br>A3-ICD-110 |
| **F-02** | **Coordinate System Mismatch** | INT-NAV-040 ↔ SRS-NAV-002<br>A3-SRS-001 |
| **F-03** | **Mass Limit Exceeded** | INT-PHY-002 ↔ IF-MEC-002<br>A3-ICD-110 |
| **F-04** | **Power Draw Exceeds Allocation** | INT-PWR-012 ↔ IF-PWR-003<br>A3-ICD-110 |

# 1. Executive summary

The incoming Interceptor Segment Specification (A3-INT-SSS-010 Rev A) was checked requirement by requirement against all 6 Arrow 3 documents uploaded so far. The check found 11 inconsistencies: 4 critical, 6 major and 1 minor. Most conflicts are with A3-ICD-110 (5).

Critical findings break a baselined interface and block the PDR baseline; major findings violate an allocated budget or environment; minor findings are documentation inconsistencies.




# 2. Conflict heat map

Where the incoming document disagrees with the baseline: each cell counts the findings for one category and one baseline document, coloured by the most severe of them.

| Category | BMC-SSS-030 | ENV-050 | ICD-110 | SRS-001 |
|---|---|---|---|---|
| Environment | · | 1 | · | · |
| Interface | · | · | 1 | · |
| Interface/Coordinates | · | · | · | 1 |
| Mass/Physical | · | · | 1 | · |
| Power | · | · | 1 | · |
| Reliability | · | 2 | · | · |
| Standards | · | 1 | · | · |
| Standards/Revision | 1 | · | · | · |
| Timeline | · | · | 1 | · |
| Timeline/Accuracy | · | · | 1 | · |

Critical Major Minor

# 3. Findings at a glance

| ID | Severity | Finding | Incoming req. | Conflicts with |
|---|---|---|---|---|
| F-01 | CRITICAL | Data Interface Mismatch | INT-DAT-020 | IF-DAT-001<br>A3-ICD-110 |
| F-02 | CRITICAL | Coordinate System Mismatch | INT-NAV-040 | SRS-NAV-002<br>A3-SRS-001 |
| F-03 | CRITICAL | Mass Limit Exceeded | INT-PHY-002 | IF-MEC-002<br>A3-ICD-110 |
| F-04 | CRITICAL | Power Draw Exceeds Allocation | INT-PWR-012 | IF-PWR-003<br>A3-ICD-110 |
| F-05 | MAJOR | Outdated Protocol Revision | INT-DAT-022 | BMC-ENG-020<br>A3-BMC-SSS-030 |
| F-06 | MAJOR | Operating Temperature Range Shortfall | INT-ENV-050 | ENV-TMP-001<br>A3-ENV-050 |

| | | | | |
|---|---|---|---|---|
| **F-07** | **MAJOR** | MTBF Shortfall | INT-RAM-070 | RAM-004<br>A3-ENV-050 |
| **F-08** | **MAJOR** | Storage Life Shortfall | INT-RAM-072 | RAM-010<br>A3-ENV-050 |
| **F-09** | **MAJOR** | Ready-Report Latency Overrun | INT-TIM-031 | IF-SEQ-006<br>A3-ICD-110 |
| **F-10** | **MAJOR** | Time Synchronization Accuracy Shortfall | INT-TIM-035 | IF-TIM-010<br>A3-ICD-110 |
| **F-11** | **MINOR** | Outdated EMC Standard | INT-EMC-060 | ENV-EMC-010<br>A3-ENV-050 |

# 4. Detailed findings

## F-01 · Data Interface Mismatch

**CRITICAL** Interface · INT-DAT-020 ↔ IF-DAT-001 (A3-ICD-110)

| INCOMING · A3-INT-SSS-010 Rev A · INT-DAT-020 | BASELINE · A3-ICD-110 · IF-DAT-001 |
|---|---|
| *"The interceptor shall exchange pre-launch data with the launcher over a dual-redundant Gigabit Ethernet (1000BASE-T) link, the interceptor acting as server."* | *"Data between the Launcher Control Unit and the AUR shall be exchanged over a dual-redundant MIL-STD-1553B bus (Bus A / Bus B)."* |

**Impact:** Complete interface incompatibility. The interceptor uses Ethernet while the launcher implements MIL-STD-1553B, preventing any pre-launch communication.

**Recommended resolution:** Align the interceptor interface with the established MIL-STD-1553B standard in A3-ICD-110.

## F-02 · Coordinate System Mismatch

**CRITICAL** Interface/Coordinates · INT-NAV-040 ↔ SRS-NAV-002 (A3-SRS-001)

| INCOMING · A3-INT-SSS-010 Rev A · INT-NAV-040 | BASELINE · A3-SRS-001 · SRS-NAV-002 |
|---|---|
| *"Pre-launch target data and in-flight target updates shall be received and processed in Earth-Centred Inertial coordinates (ECI, J2000 epoch)."* | *"All target state vectors and launch parameters exchanged between segments shall be expressed in WGS-84 Earth-Centred, Earth-Fixed (ECEF) coordinates..."* |

**Impact:** The interceptor expects ECI coordinates, but the rest of the system (BMC, FCR, Launcher) provides WGS-84 ECEF. This will result in complete navigation failure.

**Recommended resolution:** Change interceptor coordinate processing to WGS-84 ECEF.

## F-03 · Mass Limit Exceeded

**CRITICAL** Mass/Physical · INT-PHY-002 ↔ IF-MEC-002 (A3-ICD-110)

| INCOMING · A3-INT-SSS-010 Rev A · INT-PHY-002 | BASELINE · A3-ICD-110 · IF-MEC-002 |
|---|---|
| *"The mass of the all-up round, including the canister, shall not exceed 1,450 kg."* | *"The mass of the loaded canister (all-up round) shall not exceed 1,350 kg."* |

**Impact:** The interceptor exceeds the mass limit defined in the ICD and the rating of the launcher erector (LCH-MEC-014), potentially causing structural failure or inability to load.

**Recommended resolution:** Reduce interceptor mass to 1,350 kg or update launcher erector rating and ICD limits.

## F-04 · Power Draw Exceeds Allocation

**CRITICAL** Power · INT-PWR-012 ↔ IF-PWR-003 (A3-ICD-110)

| INCOMING · A3-INT-SSS-010 Rev A · INT-PWR-012 | BASELINE · A3-ICD-110 · IF-PWR-003 |
|---|---|
| *"During pre-launch (IMU run-up and thermal conditioning) the interceptor shall draw not more than 18 A continuous from the launcher supply."* | *"The current drawn by one AUR from the launcher supply shall not exceed 12 A continuous in any pre-launch state..."* |

**Impact:** The requested current (18 A) exceeds the ICD limit (12 A) and will trip the launcher's electronic circuit breakers (LCH-PWR-021).

**Recommended resolution:** Reduce power consumption to 12 A or upgrade launcher power distribution units.

## F-05 · Outdated Protocol Revision

| MAJOR | Standards/Revision · INT-DAT-022 ↔ BMC-ENG-020 (A3-BMC-SSS-030) |
|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-DAT-022 | BASELINE · A3-BMC-SSS-030 · BMC-ENG-020 |
| *"In-flight target update messages shall be decoded per A3-ICD-120 Rev A."* | *"The BMC shall transmit in-flight target updates (IFTU) to each interceptor in flight... per A3-ICD-120 Rev C."* |

**Impact:** The interceptor is specifying Rev A of the protocol, while the BMC, FCR, and Launcher are using Rev C. This may lead to message decoding failures.

**Recommended resolution:** Update the interceptor requirement to comply with A3-ICD-120 Rev C.

## F-06 · Operating Temperature Range Shortfall

| MAJOR | Environment · INT-ENV-050 ↔ ENV-TMP-001 (A3-ENV-050) |
|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-ENV-050 | BASELINE · A3-ENV-050 · ENV-TMP-001 |
| *"The all-up round shall operate at ambient temperatures from −20 °C to +50 °C."* | *"All field-deployed equipment... shall operate at ambient temperatures from −33 °C to +55 °C."* |

**Impact:** The interceptor cannot operate in the full environment specified for all field-deployed equipment, limiting deployment in extreme cold or hot climates.

**Recommended resolution:** Expand operating temperature range to -33 °C to +55 °C.

## F-07 · MTBF Shortfall

| MAJOR | Reliability · INT-RAM-070 ↔ RAM-004 (A3-ENV-050) |
|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-RAM-070 | BASELINE · A3-ENV-050 · RAM-004 |
| *"The captive-carry MTBF of the all-up round, powered on the launcher, shall be at least 1,500 h."* | *"Interceptor (all-up round) captive-carry MTBF... shall be at least 2,000 h."* |

**Impact:** The interceptor reliability is lower than the 2,000 h requirement, which may degrade the overall system operational availability (SRS-RAM-001).

**Recommended resolution:** Increase design MTBF to 2,000 h.

## F-08 · Storage Life Shortfall

| MAJOR | Reliability · INT-RAM-072 ↔ RAM-010 (A3-ENV-050) |
|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-RAM-072 | BASELINE · A3-ENV-050 · RAM-010 |
| *"The interceptor shall have a storage life of at least 10 years in its sealed canister."* | *"Interceptor storage life in the sealed canister shall be at least 15 years."* |

**Impact:** The 10-year storage life is shorter than the 15-year requirement in the baseline (RAM-010 and SRS-RAM-010), increasing the frequency of costly missile replacements.

**Recommended resolution:** Extend the sealed canister storage life to 15 years.

## F-09 · Ready-Report Latency Overrun

| MAJOR | Timeline · INT-TIM-031 ↔ IF-SEQ-006 (A3-ICD-110) |
|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-TIM-031 | BASELINE · A3-ICD-110 · IF-SEQ-006 |
| *"The interceptor shall report READY within 6.0 s of receiving the Prepare-to-Launch (PTL) command."* | *"The interceptor shall report READY (message STS, bit 0) within 3.0 s of receiving PTL."* |

**Impact:** The interceptor takes twice as long as allowed to report READY. This violates the system reaction time allocation in SRS-TL-002 and will trigger a HANG-FIRE declaration by the LCU (IF-SEQ-007).

**Recommended resolution:** Reduce the READY report time to 3.0 s.

### F-10 · Time Synchronization Accuracy Shortfall

| MAJOR | Timeline/Accuracy · INT-TIM-035 ↔ IF-TIM-010 (A3-ICD-110) | |
|---|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-TIM-035 | | BASELINE · A3-ICD-110 · IF-TIM-010 |
| *"The interceptor time reference shall be synchronised to system time within ±10 µs."* | | *"the interceptor shall align its time reference within ±1 µs."* |

**Impact:** The synchronization accuracy is an order of magnitude worse than required by the ICD (IF-TIM-010) and SRS (SRS-TIM-001), potentially degrading target track precision.

**Recommended resolution:** Improve clock synchronization accuracy to ±1 µs.

### F-11 · Outdated EMC Standard

| MINOR | Standards · INT-EMC-060 ↔ ENV-EMC-010 (A3-ENV-050) | |
|---|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-EMC-060 | | BASELINE · A3-ENV-050 · ENV-EMC-010 |
| *"The interceptor shall meet MIL-STD-461F requirements CE102, CS114, RE102 and RS103."* | | *"All segments shall meet MIL-STD-461G requirements CE102, CS101, CS114, RE102 and RS103 as tailored in Annex B."* |

**Impact:** The interceptor specifies Revision F while the system baseline requires Revision G. This may lead to non-compliance with the latest EMC requirements.

**Recommended resolution:** Update EMC requirements to MIL-STD-461G.

## 5. Action plan

Proposed actions for the next Interface Working Group and Configuration Control Board.

| ID | Action | Owner | Due |
|---|---|---|---|
| F-01 | Align the interceptor interface with the established MIL-STD-1553B standard in A3-ICD-110. | Interface Working Group (INT/LCH) | Before PDR data package |
| F-02 | Change interceptor coordinate processing to WGS-84 ECEF. | Interface Working Group (INT/LCH) | Before PDR data package |
| F-03 | Reduce interceptor mass to 1,350 kg or update launcher erector rating and ICD limits. | Interface Working Group (INT/LCH) | Before PDR data package |
| F-04 | Reduce power consumption to 12 A or upgrade launcher power distribution units. | Interface Working Group (INT/LCH) | Before PDR data package |
| F-05 | Update the interceptor requirement to comply with A3-ICD-120 Rev C. | Interceptor IPT | PDR |
| F-06 | Expand operating temperature range to -33 °C to +55 °C. | Specialty Engineering | PDR |
| F-07 | Increase design MTBF to 2,000 h. | Specialty Engineering | PDR |
| F-08 | Extend the sealed canister storage life to 15 years. | Specialty Engineering | PDR |
| F-09 | Reduce the READY report time to 3.0 s. | Weapon System Engineering | PDR |
| F-10 | Improve clock synchronization accuracy to ±1 µs. | Weapon System Engineering | PDR |
| F-11 | Update EMC requirements to MIL-STD-461G. | Interceptor IPT | Next revision |

## Appendix A. Baseline documents checked

| Document | Rev | Title | Requirements |
|---|---|---|---|
| A3-BMC-SSS-030 | A | Battle Management Center Specification | 7 |
| A3-ENV-050 | B | Environmental and RAM Requirements | 10 |
| A3-FCR-SSS-020 | B | Fire Control Radar Specification | 8 |
| A3-ICD-110 | C | Interceptor–Launcher ICD | 13 |
| A3-LCH-SSS-040 | B | Launcher Segment Specification | 13 |
| A3-SRS-001 | C | System Requirements Specification | 18 |

# Appendix B. How this report was produced

1. Trigger: A3-INT-SSS-010 Rev A was dropped onto the agent in EDK, waking both Hermes agents.
2. Baseline register (main Hermes): index/arrow3-baseline.md holds every requirement of every other Word document uploaded so far; only new or changed documents are re-read.
3. Spec Reader (second Hermes, in parallel): converted the incoming document to Markdown with pandoc, extracted every "shall" statement and wrote a management briefing.
4. Main Hermes studied the register and the new requirement lines together, then compared every incoming requirement with the baseline.
5. This report was generated from the findings by the program report kit (consistency_report.py).

*Automated findings support, but do not replace, engineering review. Confirm each finding against the source documents before CCB submission.*